

# Image Processing – 67829 – Exercise 5

**Due Date: 22.01.2026 at 23:59**

**Version 1.0 - Last Update January 8, 2026**

**Note:** You will be implementing the exercise using Google Colab. Colab is an IPython notebook running on Google cloud servers and provides you with GPU. Please make sure to follow the instructions provided in the following [Colab Notebook](#) (Use HUJI user - open in Colab). As neural networks use GPU for fast and efficient computations, it is strongly advised you make sure you are training using a GPU and not a CPU, more on this in the notebook.

**Note 2:** When initially opening the notebook there should be a text to the right of the "Help" menu saying "Changes will not be saved". To ensure you can make changes to the notebook save a copy of it to your own drive and work on that one. You can do that by going to "File" -> "Save a copy in Drive".

**Failing to do so will result in code loss!**

**Make sure you are the only one that has access to it!**

**Note 3:** This exercise is designed to run entirely on the free tier, and it is possible to complete it using only the free tier. However, the free tier has a daily limit on GPU usage. If this proves insufficient, you may choose to purchase Colab Premium, which significantly increases these usage limits.

## 1 Overview

In this exercise, you will explore Stable Diffusion (SD) as a generative image prior. You will first "teach" a model a new concept using a simplified version of DreamBooth [3] with LoRA [1]. DreamBooth works by fine-tuning the model to associate a set of images of a specific subject with a unique text identifier. This process allows the model to "memorize" the identity and details of your subject so it can be reconstructed in novel contexts using text prompts. You will then implement and compare different editing techniques: text-based prompting and SDEdit [2]. SDEdit facilitates image editing by adding a controlled amount of noise to an existing image and then denoising it guided by a new text prompt. This allows for structural changes while maintaining the global composition and spatial layout of the original input.

**Note:** To allow you to work on the exercise quickly, and without long training times or large GPUs, we will be using an older version of Stable Diffusion (Stable Diffusion 1.5). However, the ideas and solutions can be easily adopted to more recent diffusion models for better quality results.

## 2 Tasks

### 2.1 Personalization (Simplified DreamBooth)

Your first task is to fine-tune a pre-trained Stable Diffusion model to associate a specific subject (e.g., your pet or a unique object) with a unique text token (e.g., `my_token`). Fine-tuning involves training the model so that when it sees the token `my_token`, it generates your specific subject rather than a generic version of that class.

**Note:** In the notebook we provide an example dataset for getting started. In your submission you must use your own datasets and can't use this dataset.

### 2.2 Personalized Image Generation and Editing

You will use the **SDEdit** algorithm, which involves adding noise to an image and denoising it using a prompt as a guide. You will then compare three distinct methods for generating personalized images or editing existing ones:

1. **SDEdit (Generic):** Using the base SD model with no fine-tuning.
2. **DreamBooth (Fine-tuned):** Using your personalized LoRA model with pure text prompting.
3. **DreamBooth + SDEdit:** Using your personalized LoRA as the backbone for an SDEdit guided reconstruction.

### 2.3 Forensics Challenge (Membership Inference)

In Moodle, under “Exercise 5” -> “**Ex5 Forensics Supplementary**”, you are given a fine-tuned model and 6 images. Three of these images were part of the training set used for fine-tuning, while other three are new images of the same person. Your task is to develop and implement an algorithm to detect which images were used during the fine-tuning process. Your algorithm should take in an image and return a score that can be used to determine whether the image was used for training. When using the algorithm with a number of images, some from the training dataset and some not from it, we should be able to separate the scores of the training samples from the non-training samples.

**Tip:** Create a similar setup using a model you’ve trained so that you can play with it while knowing which images were used for training.

**Note:** You will most likely need to choose a threshold to determine which images were used for training. For the purpose of this exercise, you can decide on this threshold by clustering your results (the scores from the algorithm) into two groups and selecting a value that separates these clusters effectively. In practice, metrics like ROC AUC are often used to choose such thresholds.

## 3 Submission

Submission instructions are given in the “Submissions Guidelines” document published on the course web page ([here](#)). Please read and follow them carefully. Any updates to those guidelines will be posted in the news forum.

**You must submit your code but there is no API you need to follow and there are no presubmissions.**

### 3.1 Report Guidelines

In addition to the code, you should submit a report describing your solution. The report must follow the following structure and address the topics below. We provide an [English](#) and [Hebrew](#) Google Docs template (you need to copy it to use it). In case you choose not to use it, please maintain a similar structure. The report should be no longer than 6 pages and include the following sections:

1. Introduction and Theory

- (a) In your own words, state the goal of the exercise.
- (b) Provide a simple explanation of how a diffusion model works and what the process of fine-tuning entails.

2. Algorithm and Implementation

- (a) **SDEdit:** Clearly describe the conceptual steps of adding noise and denoising for guided editing.
- (b) **Forensic Methodology:** Describe the logic and steps of your Membership Inference solution. Describe the criteria used to distinguish training images from unseen ones.

3. Results and Analysis

- (a) **Training Progression:** Show results throughout the training process that demonstrate the transition from underfitting to convergence to overfitting (include 2 examples of underfitting, 2 of a converged model, and 2 of overfitting). Explain the concepts of underfitting and overfitting in this context and why your chosen images demonstrate these states. **Note:** You do not need multiple training runs.

Instead, you should save *intermediate checkpoints* throughout a single training session to observe the progression from underfitting to convergence and finally overfitting.

- (b) **Dataset Comparison:** Explain the characteristics that distinguish a “good” dataset from a “bad” dataset when training these DreamBooth personalization models. Train a model on both a good and a bad dataset and present the training samples of both and 2 generated results from each to support your claims (you may use the same hyper-parameters for both runs).
  - (c) **SDEdit Variations:** Show results of SDEdit without fine-tuning, demonstrating how different amounts of noise affect the balance between the original image and the text prompt (show the original image and results from 3 different noise levels). Briefly discuss why the noise level affects the results.
  - (d) **Method Comparison:** Compare the different methods (**SDEdit**, **DreamBooth**, and **DreamBooth + SDEdit**) by presenting 3 images generated from each. Analyze the pros and cons of each method, discussing specific use cases for each and their respective failure modes.
4. Forensics
- (a) Present the results of your detection algorithm (e.g., using figures that explain your results and plots that show the different scores) that support your findings.
  - (b) List the images (the file names) you’ve identified as training samples.
5. Conclusion
- (a) Summarize your key findings and insights from the exercise.

**Note: For each generated or edited image, also include the prompt you’ve used.**

Your final submission should be:

1. A PDF named “ex5.pdf”.
2. A tar file containing your `.ipynb` notebook and all input/output images used in your report (with meaningful file names). To create a tar file you can run the following command: `tar -cvf ex5.tar ex5.ipynb images/`

**Note: The PDF and tar should be submitted to the respective submission in the Moodle.**

## 4 Grading

Your exercise will be graded based on a manual inspection of your report (and code). There will be no pre-submission tests and no automatic tests.

## References

- [1] Edward J Hu, Yelong Shen, Phillip Wallis, Zeyuan Allen-Zhu, Yuanzhi Li, Shean Wang, Lu Wang, Weizhu Chen, et al. Lora: Low-rank adaptation of large language models. *ICLR*, 1(2):3, 2022. [1](#)
- [2] Chenlin Meng, Yutong He, Yang Song, Jiaming Song, Jiajun Wu, Jun-Yan Zhu, and Stefano Ermon. SDEdit: Guided image synthesis and editing with stochastic differential equations. In *International Conference on Learning Representations*, 2022. [1](#)
- [3] Nataniel Ruiz, Yuanzhen Li, Varun Jampani, Yael Pritch, Michael Rubinstein, and Kfir Aberman. Dream-booth: Fine tuning text-to-image diffusion models for subject-driven generation. In *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition*, pages 22500–22510, 2023. [1](#)